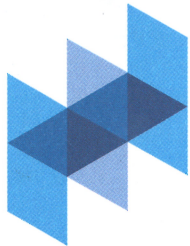


SOLUTION



HÖGSKOLAN
I HALMSTAD

①

	Security	Traceability	Complexity
a) Passw.	-	-	+
Biometrics	+	+	-

• passw: easy to copy, share, steal, guess, lose...

• biom: cannot be lost, forgotten; more difficult to copy, steal

• passw: not linked to identity, those who have/know it have the same privilege

• biom: unique, linked to the person

• passw: easy & cheap

• biom: more expensive & complex

b) Verification: higher security than identification, better if you know the list of "authorized people"

Modality: for high security: fingerprint or iris*
(better than for example face or voice)

Database: local (chip card with fingerprint stored in it)

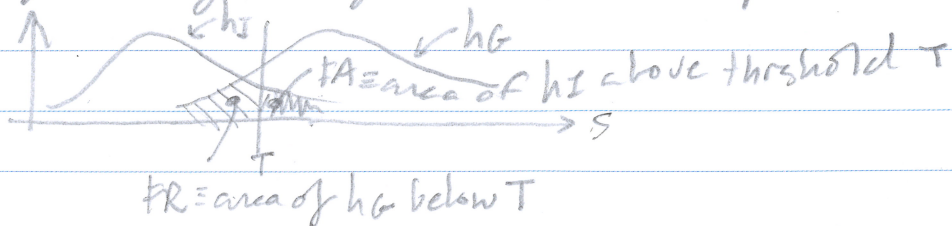
↳ more secure (to avoid data travelling through the network)

↳ comparison/matching done in the card or in the sensor, only decision is given

↳ identification mode is not possible with local database

c) Carry out comparisons (matchings) with a group of controlled genuine/impostor persons

Plot histograms of h_g (genuine), h_i (impostors)



Compute the two types of errors FAR/FRR for different thresholds T

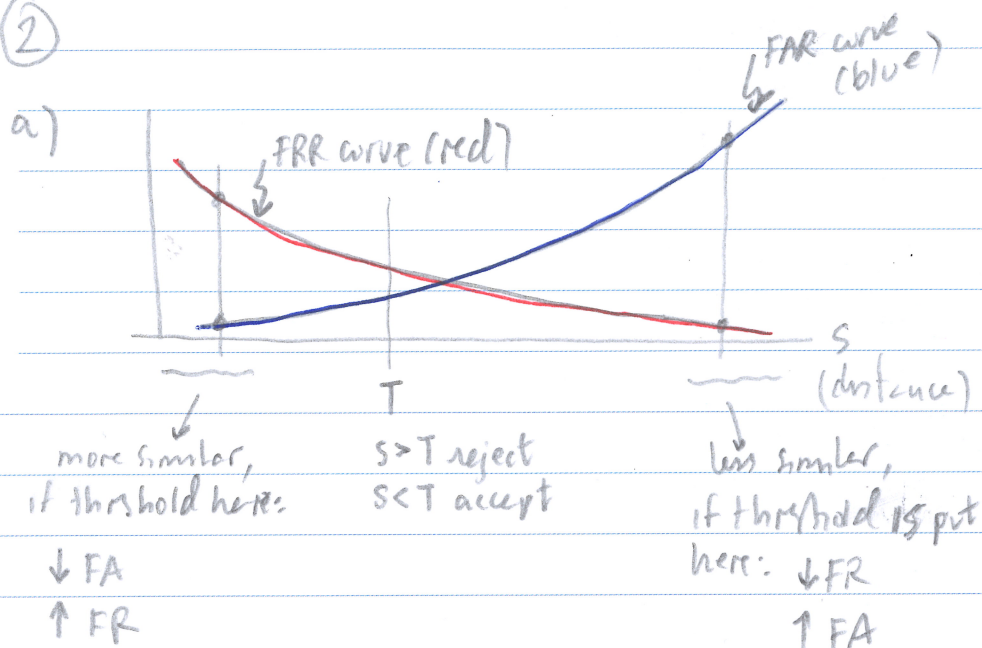
$S > T$ accept / $S < T$ reject



HÖGSKOLAN
I HALMSTAD

②

a)



b) High Security $\rightarrow$ Low FAR

Choose smaller T

To avoid $FRR > 10\%$, then T should be > 5

At the same time, to have low FAR:

T = 7 : FAR $\sim 1\%$, FRR $\sim 4\%$

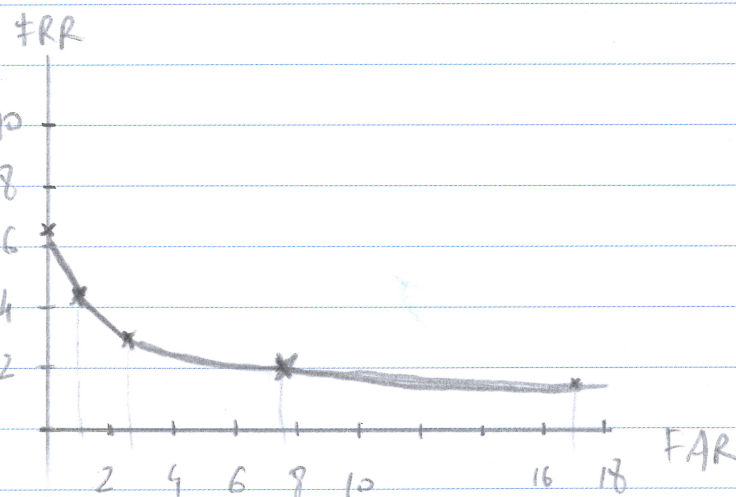
T = 6 : FAR $\sim 0\%$, FRR $\sim 6-7\%$

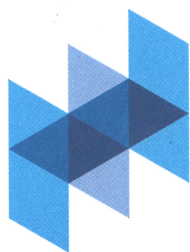
choice of T

between 6-7

c)

T	FRR (red)	FAR (blue)
4	16	0
5	10.5	0
6	6.5	0
7	4.5	1
8	3	2.5
9	2	7.5
10	1.5	17





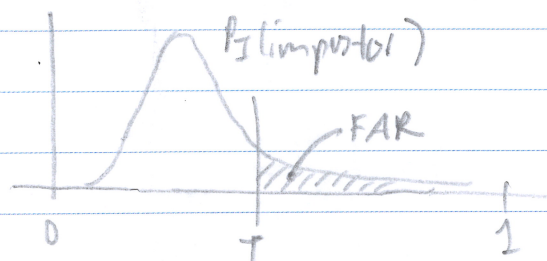
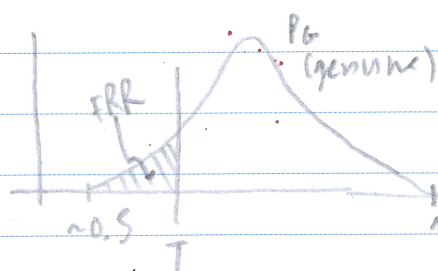
HÖGSKOLAN
I HALMSTAD

③ verification mode, $\begin{cases} 1200 \text{ genuine matchings} \\ 1800 \text{ impostor} \end{cases}$

a) Similarity measure, $S > T$ accept
 $S \leq T$ reject

$$FRR(T) = \sum_{S=0}^T P_G(S)$$

$$FAR(T) = \sum_{S=T}^1 P_I(S)$$

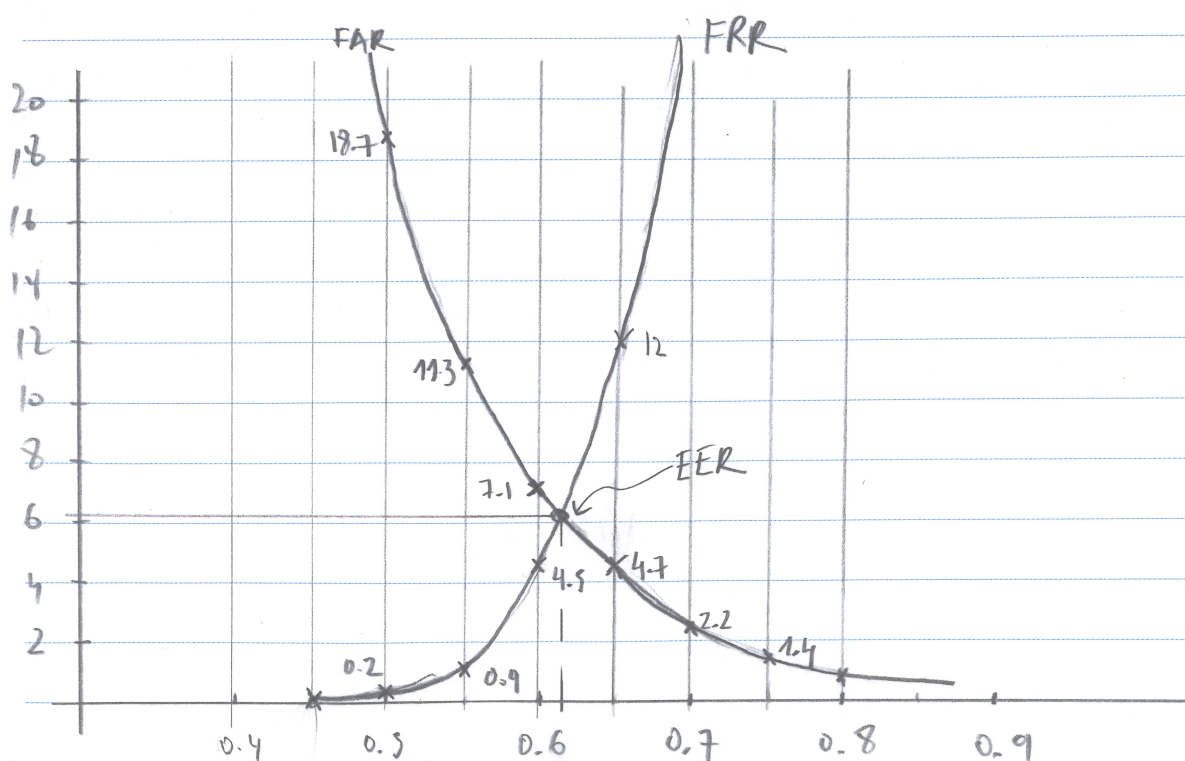


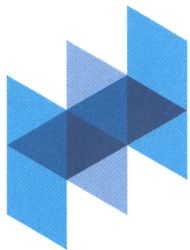
- overlap is from $T \approx 0.5$

% of genuine matchings

% of impostor matchings

T	genuine (amount $\leq T$)	impostor (amount $> T$)	FRR (% below T)	FAR (% above T)	T
0.45	0	448	$0/1200 = 0 = 0\%$	$448/1800 = 0.249 = 24.9\%$	0.45
0.5	2	337	$2/1200 = 0.002 = 0.2\%$	$337/1800 = 0.187 = 18.7\%$	0.5
0.55	11	203	0.9%	11.3%	0.55
0.6	54	127	4.5%	7.1%	0.6
0.65	144	85	12%	4.7%	0.65
0.7	337	40	28.1%	2.2%	0.7
0.75	607	26	50.6%	1.4%	0.75
0.8	895	16	74.6%	0.9%	0.8





HÖGSKOLAN
I HALMSTAD

b) From the figure: EER $\approx 6-6.1\%$
at $T \approx 0.61-0.62$

c) FRR = 10% is at $T \approx 0.64$

At this point FAR $\approx 4.8\% > 2\%$.

So the system cannot meet the requirement

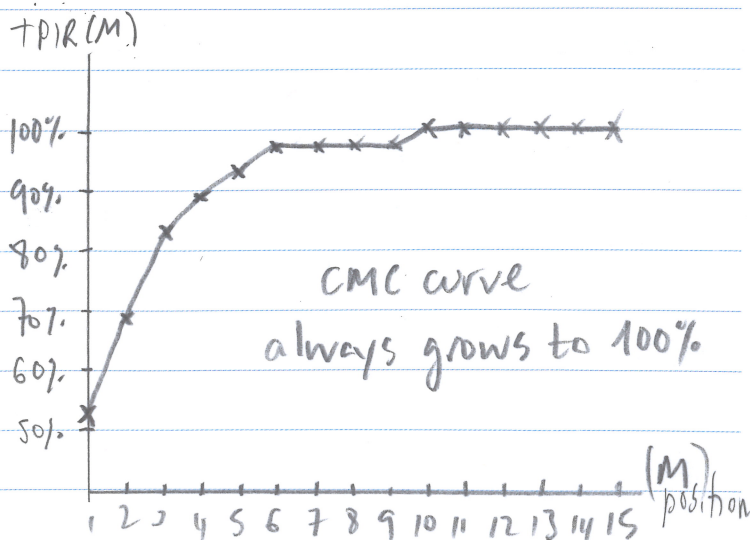
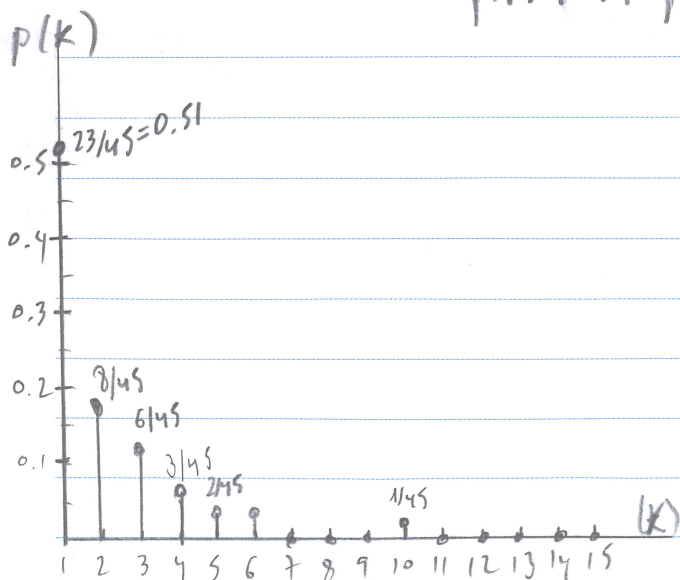
4) Identification mode, 15 persons, $15 \times 3 = 45$ tests in total

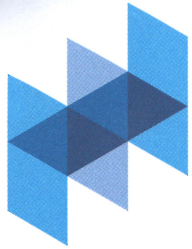
a)

Position (k)	1	2	3	4	5	6	7	8	9	10	11	...	15
# of correct ID	23	8	6	3	2	2	0	0	0	1	0	...	0
P(k)	$\frac{23}{45}$ (0.51)	$\frac{8}{45}$ (0.178)	$\frac{6}{45}$ (0.13)	$\frac{3}{45}$ (0.07)	$\frac{2}{45}$ (0.04)	$\frac{2}{45}$ (0.04)	0	0	0	$\frac{1}{45}$ (0.02)	0	...	0
TPIR(M)	$\frac{23}{45}$ 0.51 51%	$\frac{23+8}{45}$ 0.69 69%	$\frac{23+8+6}{45}$ 0.822 82.2%	...	...	...	...	...	...	...	...	...	...

$P(k)$ = probability that the correct ID is in position k of the output list

$TPIR(M) = \sum_{k=1}^M P(k)$ = probability that the correct ID is in the first M positions of the output list





HÖGSKOLAN
I HALMSTAD

5

$$b) M=3 \Rightarrow \text{TPIR}(3) = 82.2\% < 85\%$$

the system does not meet the
requirement